

# Edwin Calles Molina

Aerospace Engineering student at Cal Poly Pomona. I design rockets, build them, fly them, and work from the data that comes back.

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**4,135 ft**  
Level 2 certification flight apogee

**Mach 1.2**  
peak speed on my second rocket

**4**  
documented flight tests on 2 vehicles

**3.97**  
GPA · President's Honor List

I'm a second-year Aerospace Engineering student at Cal Poly Pomona with a 3.97 GPA, NAR Level 2 High Power Rocketry certification, and four documented flight tests behind me. I designed these rockets in SolidWorks, 3D-printed and assembled the hardware, assembled and operated the data acquisition and GPS, and flew them.

On my most recent flight I reconstructed a supersonic structural failure from the recovered flight record, caught a corrupted altimeter reading in the process, validated the real trajectory against backup inertial data, and specified the redesign that followed.

## FLIGHT LOG

| FLIGHT | DATE         | VEHICLE            | APOGEE                | MAX VELOCITY          | OUTCOME                                                                                                                                                         |
|--------|--------------|--------------------|-----------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1      | Jul 18, 2026 | Rocket 1           | 4,224 ft              | 659 ft/s              | <div>Failure</div> Level 2 attempt failed. Descent rate went from 44 to 46 ft/s at main deployment instead of slowing. Fins damaged. Recovery train redesigned. |
| 2      | Aug 8, 2026  | Rocket 1 (rebuilt) | 4,135 ft              | 658 ft/s              | <div>Nominal</div> NAR Level 2 certification earned on a J250 motor (~700 N·s). 6.8° tilt at burnout.                                                           |
| 3      | Aug 8, 2026  | Rocket 2           | 2,833 ft              | 481 ft/s              | <div>Nominal</div> Recovery nominal, both events on time. Flight computer supply dipped to 0.31 V for about a second at apogee.                                 |
| 4      | Aug 15, 2026 | Rocket 2           | Breakup at ≈ 1,800 ft | 1,367 ft/s (Mach 1.2) | <div>Failure</div> Structural failure at t = 2.20 s. Fin failure identified as the initiating event. Redesign specified.                                        |

## IN THIS PORTFOLIO

|                                                              |                                                     |
|--------------------------------------------------------------|-----------------------------------------------------|
| Dual-deploy high power rocket — NAR Level 2 certification    | INDEPENDENT PROJECT · ROCKET 1 · JUNE – AUGUST 2026 |
| Supersonic flight test and failure reconstruction            | INDEPENDENT PROJECT · ROCKET 2 · AUGUST 2026        |
| Project EXO Bronco — Electronics & Launch Operations co-lead | UMBRA · CAL POLY POMONA · MAY 2026 – PRESENT        |
| Small Rocket Competition — sub-team lead                     | UMBRA · CAL POLY POMONA · SEPTEMBER – DECEMBER 2025 |
| SR-30 turbojet performance test                              | CAL POLY POMONA · FALL 2025                         |
| Flight log · Now · Skills · About                            | REFERENCE                                           |

U.S. citizen · eligible for a U.S. government security clearance. Available June 2027 · Summer 2027 engineering internship · willing to relocate · onsite five days a week. Every number here comes from a flight log, a lab report, or a document I wrote for a team; flight figures are from the Featherweight Blue Raven 2.0.

# Dual-deploy high power rocket

## – NAR Level 2 certification

3.1 in Blue Tube airframe · 54 mm motor mount · AeroTech J250W, 709 N·s · 3D-printed PETG ogive nosecone · birch plywood clipped-delta fins, 4.30 in root chord · dual-deploy recovery, 48 in main · Featherweight Blue Raven 2.0 · Eggfinder GPS · two flights

**ROLE**

Designer, builder, and test operator. I designed the vehicle and the electronics bay, wired and soldered the recovery electronics, and flew it.

**STATUS**

Certified · NAR Level 2 High Power Rocketry, August 2026

**TOOLS**

SolidWorks · OpenRocket · FDM 3D printing · Soldering and wiring harness assembly · Dual-deploy recovery design · Blue Raven 2.0 altimeter setup · Ejection charge sizing · Flight data analysis

**4,135 ft**

apogee on the certification flight

**6.8°**

tilt from vertical at burnout

**1,498 ft**

main deployment · programmed for 1,500 ft

**19 ft/s**

descent rate under the main

**59.6 G**

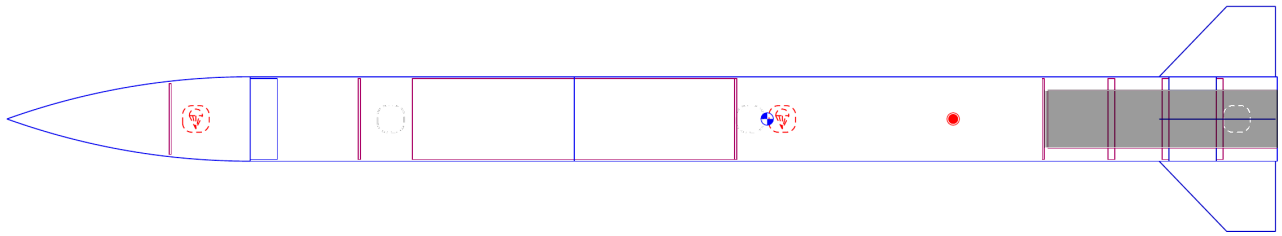
touchdown load the printed bay survived

My first Level 2 attempt failed right in front of me: the drogue came out, the main never did. I found the cause in the recovery train, restaged it, and flew again. The second flight went to 4,135 feet and certified me.

### THE OBJECTIVE

Since the end of Fall 2025, when I launched my first ever rocket as part of the Small Rocket Competition, I've been working on my High Power Rocketry certifications. By the end of spring semester I had my Level 1 certification, and over the summer I set my sights on Level 2. A 4x increase in motor power as compared to my Level 1 certification, on top of incorporating electronics that were completely new, meant setbacks were plenty. I also wanted to step up to a dual-deploy system where two parachutes are used instead of one — one parachute deploying at apogee and the other at 1,500 feet.

The Level 1 rocket I certified on that spring was a smaller vehicle, a 3.2 inch cardboard airframe on a 38 mm AeroTech H100W, and the rule I set myself was that nothing on it would be a kit part. I drew the four fins and the two centering rings and cut them on the laser cutter on campus, printed the nosecone in PLA, and built it over about two weeks in April. It flew and I certified on it, but it carried no altimeter and a single parachute on motor ejection, so every number I have for that flight is a prediction rather than a measurement. That gap is the reason this rocket carried a Blue Raven and a GPS tracker, and the reason there is any data below at all.

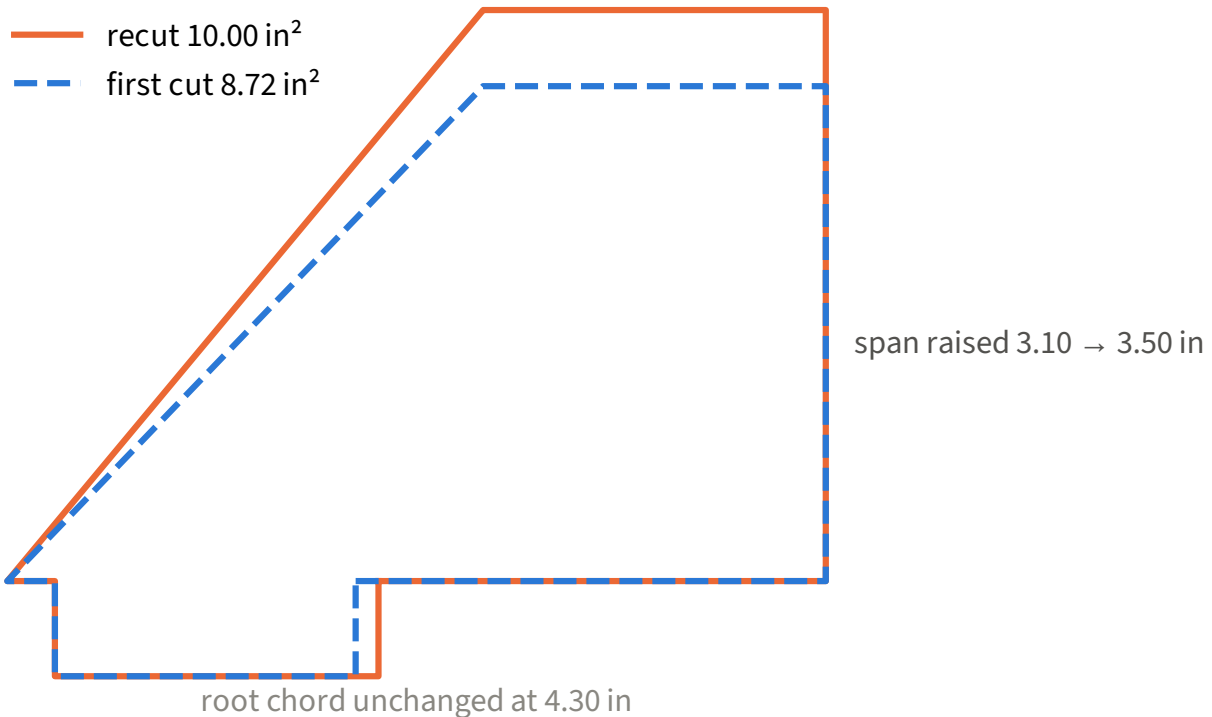


The certification vehicle in OpenRocket: 9 in printed PETG nosecone, 12 in upper body tube over the coupler and avionics bay, main chute ahead of it, and the 26 in lower body tube on a 54 mm motor mount. The centre of gravity sits 2.2 calibers ahead of the centre of pressure.

### SIZING THE FINS

I cut a first set of fins, then went back to the simulation and found the stability margin sitting lower than I wanted, which meant the fins were simply too small for the airframe. I kept the root chord at 4.30 inches and raised the span from 3.10 to 3.50 inches, taking the fin area from 8.72 to 10.00 square inches. That is a 15 percent increase in area, and it brought the static margin up near the 2 calibers I was aiming for. Both sets came off the laser cutter on campus and I still have the drawings for both, which is the only reason I can put a number on the change.

### Rocket 1 fin: first cut, then recut 15% larger for stability



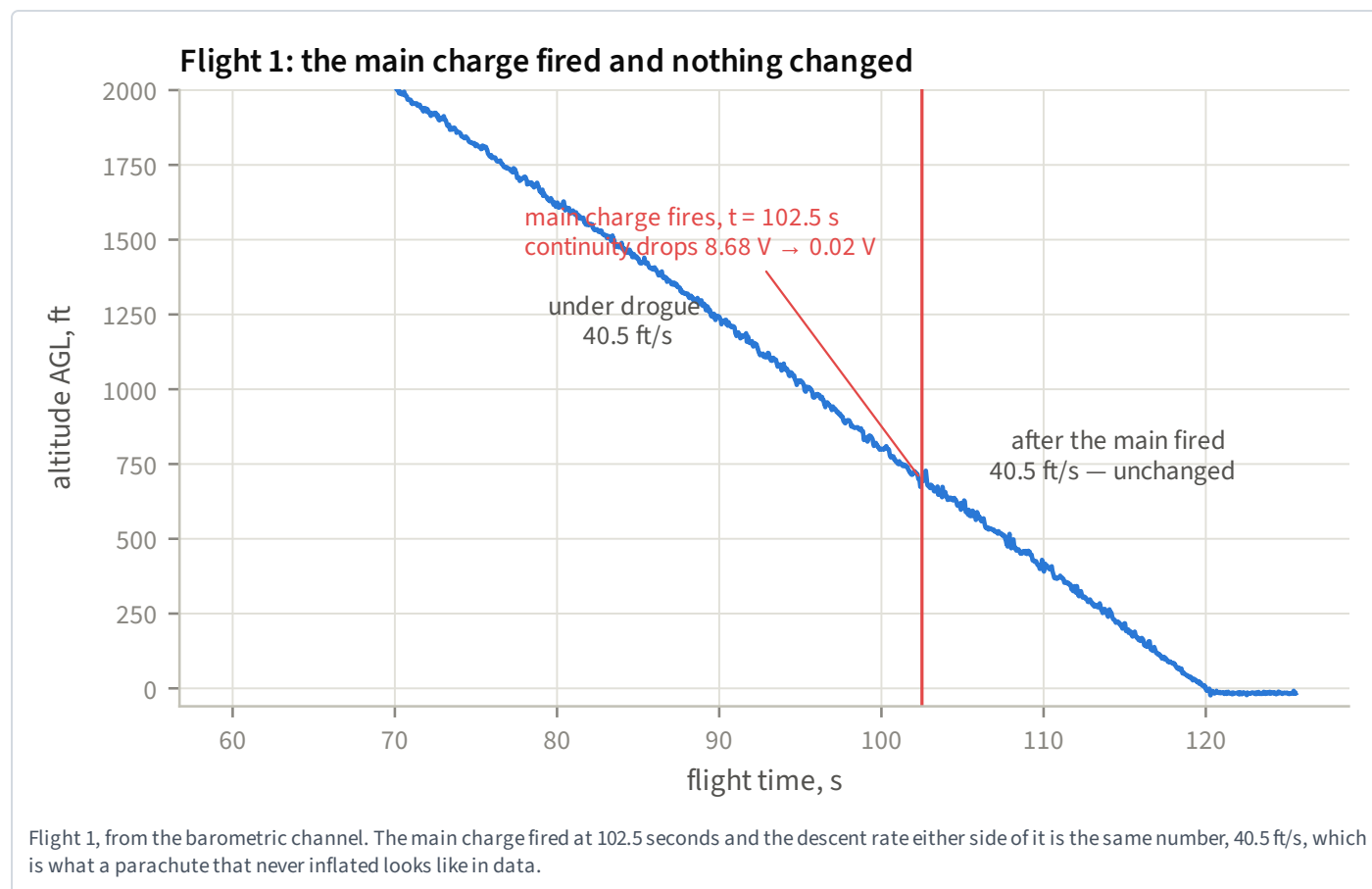
The Rocket 1 fin as first cut and as recut, taken straight from the two laser-cutter drawings. Same 4.30 in root chord, span raised from 3.10 to 3.50 in, which is 15 percent more area and the stability margin I needed.

### FLIGHT 1 – THE FAILURE

Over time, I designed it, built it, and flew it. The main chute never opened. An incorrect separation during recovery prevented it from deploying at its programmed altitude. Luckily, the drogue chute worked — enough to save the airframe, but not the fins. I was frustrated. Half my summer was sitting around me in pieces.

The rocket flew on 18 July 2026 at the Lucerne Valley lakebed with the Rocketry Organization of California, on the same AeroTech J250W I would later certify on, and it reached 4,224 feet. The flight record says the rest of it plainly. The main charge fired where it was supposed to, at 102.5 seconds and 694 feet, and I can see it fire because the continuity voltage on that channel drops from 8.68 volts to 0.02 volts inside a single 20 millisecond sample, which is the e-match burning through. What did not change was the descent rate. The rocket was coming down at 40.5 feet per second under the drogue before the charge fired, and it was still coming down at 40.5 feet per second afterwards, meaning the canopy never inflated and the vehicle rode the drogue the whole way to the ground.

It landed hard enough to break the fins. One came off the airframe entirely, a second was damaged, a third cracked at the root, and the fourth was untouched.

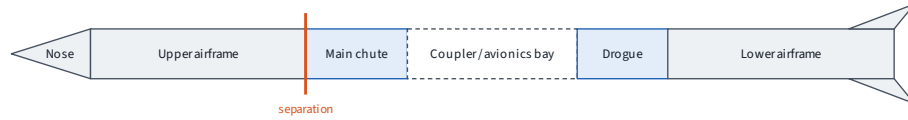


#### FINDING THE CAUSE

So I did the only thing an engineer would do. I took it home, spent weeks analyzing the data and wreckage, and when I believed I had figured out the issue, redesigned it, and flew it again. The separation point for the main parachute was incorrectly placed. The first time around I had accounted for the length of the shock cord but not for the shroud lines attached to the parachute. I believed that the shock cord alone would be enough to pull the main chute out of the airframe, but the shroud lines got pulled first, and their length acted to disperse the pull on the parachute, so the canopy never cleared the upper airframe, which was still attached to the nosecone.

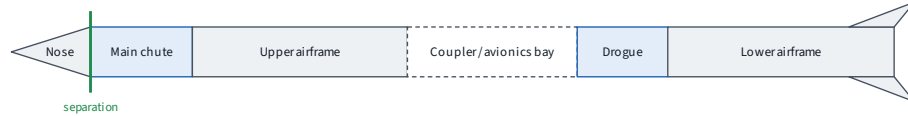
The fix was not a stronger part. I restaged the recovery train so that the main parachute sits directly behind the nosecone: at separation the nosecone pops off and frees the canopy directly.

### Flight 1 — main did not deploy



At the main event the upper airframe and nosecone separated from the coupler as one piece. The shock cord pulled the shroud lines, but the canopy stayed inside the upper airframe.

### Flight 2 — restaged, certified at 4,135 ft



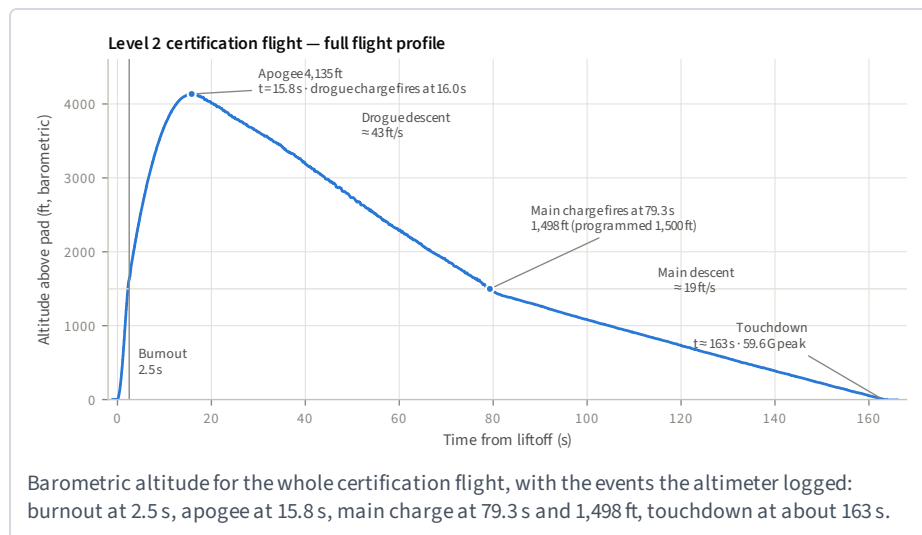
The main chute now sits directly behind the nosecone. At the main event the nosecone pops off and pulls the canopy out directly.

How the recovery train changed between the two flights. Schematic, not to scale.

## FLIGHT 2 — CERTIFICATION

That second flight is the one that certified me: a J250 motor to 4,135 feet, main deployment at 1,498 feet, and a descent rate of 19 ft/s, right on target. As an additional testament to the design and construction of the rocket, the rocket stayed within 6.8 degrees of vertical at burnout, meaning that the rocket flew straight up until burnout. The recovery system did exactly what I designed it to do. The main parachute deployed at 1,498 feet, the altitude I had programmed it to, and brought the rocket down at 19 feet per second.

The electronics bay I designed and 3D-printed protected the altimeter and GPS through launch, both ejection charges, and a 59.6 G touchdown, and the GPS tracker maintained tracking through recovery.



## HOW CLOSE THE PREDICTION WAS

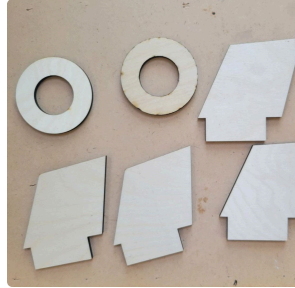
I had simulated the flight in OpenRocket before I flew it, and afterwards I could check the prediction against the record. The model said 4,561 feet and 705 feet per second, and the rocket flew 4,135 feet and 658 feet per second, so I was 9 percent high on altitude and 7 percent high on speed. The landing velocity it predicted was 19.9 feet per second against the 19 I measured. I had not thought of it as a prediction-then-check exercise at the time, but that is what it was, and it is the closest I have come so far to knowing how a vehicle would behave before building it.

## WHAT I LEARNED

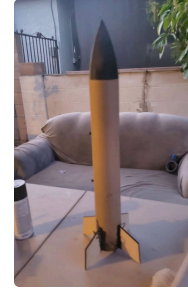
This failure taught me to be more thorough in my designing but more specifically my testing. This kind of mistake could have been avoided through more rigorous testing and by verifying that the parachute gets pulled out during separation testing on the ground. All in all, what has stayed with me is this: failure is a lesson waiting to be discovered. I prepared what I could, but it was watching the failure of my rocket that taught me the final lesson.



The certification flight leaving the pad, 8 August 2026. Frame from my own video of the launch. This is the flight that reached 4,135 ft and earned the Level 2.



The Level 1 parts I drew and cut on the laser cutter on campus, 27 March 2026: two centering rings and four fins in 0.25 in birch plywood. The step at the bottom of each fin is the through-the-wall tab.



The finished Level 1 vehicle, 10 April 2026. It carried no altimeter, which is why I have a prediction for that flight and not a measurement.

# Supersonic flight test and failure reconstruction

Scratch-built 3.1 in Blue Tube airframe, 82 in long · 54 mm fiberglass motor mount · fiberglass ogive nosecone · fiberglass clipped-delta fins, 5.00 in root chord, 0.25 in thick · dual-deploy recovery · Featherweight Blue Raven 2.0 · Eggfinder GPS · two flights

|        |                                                                                                                                                                                                |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROLE   | Sole designer, builder, test operator and analyst. The airframe, fins and electronics bay are my design and my build, the altimeter is off the shelf, and the tracker is a kit I soldered.     |
| STATUS | Redesign specified · rebuild in progress · not yet re-flown                                                                                                                                    |
| TOOLS  | SolidWorks · OpenRocket · FDM 3D printing · Soldering (Eggfinder GPS / TX / RX kits) · Blue Raven 2.0 altimeter setup · Ejection charge sizing · Flight data reduction · Failure investigation |

**1,367 ft/s**  
peak vertical velocity  
· ≈ Mach 1.2

**2.20 s**  
time from liftoff to breakup

**3 → 88 G**  
side load growth in ≈ 0.35 s

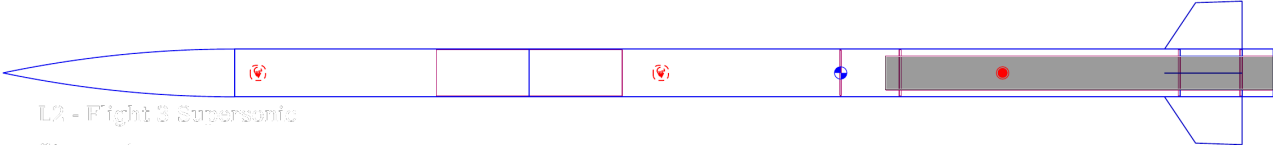
**535 → 290**  
roll rate decay, deg/s, before rupture

**500 Hz**  
inertial record recovered intact

My second rocket reached Mach 1.2 and broke apart in flight. I recovered the electronics bay intact, pulled the full 500 Hz flight record off it, and spent weeks working out from the data and the wreckage that the fins failed first and the airframe opened afterward.

## THE OBJECTIVE

After earning my Level 2 certification, I wanted to build a rocket that could go past Mach 1. I designed it from scratch around a 3.1 inch Blue Tube airframe and a 54 mm fiberglass motor mount, with a dual-deploy recovery system and a custom 3D-printed electronics bay holding the Blue Raven altimeter and the Eggfinder GPS tracker. I checked the design in OpenRocket, which put the static margin at 3.37 calibers and predicted 1,497 ft/s on an AeroTech L1000W, sized the ejection charges for the airframe volume, and sourced custom G10 fiberglass centering rings for the airframe.



The vehicle as I drew it in OpenRocket: fiberglass ogive nosecone, 19 in upper body tube, the coupler and avionics bay, a 48 in lower body tube over a 54 mm fiberglass motor mount, and four fins. The blue quartered marker is the centre of gravity and the red one the centre of pressure, 3.37 calibers apart.

## THE STAGED APPROACH

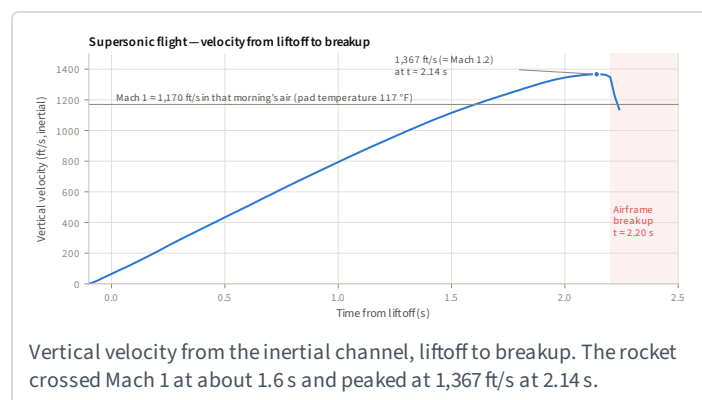
Rather than flying the rocket straight to its design limit, I flew it on a smaller motor first as a subsonic shakedown. That flight reached 2,833 feet, separated at apogee, and deployed the main parachute at 1,499 feet for an 18 ft/s descent, which confirmed that the airframe, the recovery system, and the electronics all worked before I risked the vehicle on a supersonic attempt.

There is one thing in the shakedown record I already knew about and want stated rather than left in a CSV for someone else to find. The flight computer runs off an ordinary 9 volt alkaline, the kind sold for smoke alarms, which also feeds the magnetic arming switch, and the Eggfinder transmitter runs off a separate two cell LiFe pack. When the drogue charge fired at apogee the supply to the flight computer dropped to 0.31 volts for about a second. That is close to a total collapse rather than the ordinary sag an alkaline cell shows under an e-match load, and I do not have an explanation for it that I am willing to put my name to yet. I fit a fresh 9 volt for every flight, and the certification flight a week earlier, on the same setup and its own fresh cell, never went below 8.9 volts. Both deployment events still fired on time and the vehicle came back clean, so I flew it again as it was, but the power path is the first thing I am instrumenting on the rebuild.

Before either flight I ground-tested the ejection charges in the actual airframe, on 22 July, with the section resting on a stand and cushions set downrange to catch it. Four tests, all successful. Sizing a charge from the airframe volume gives you a number, but firing it in the real tube is what tells you whether that number is right.

#### WHAT HAPPENED

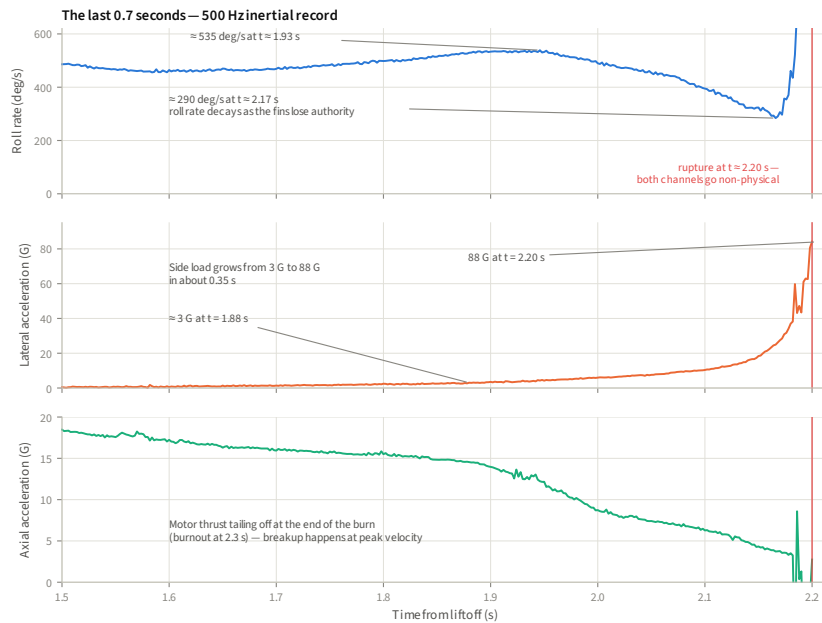
One week later I flew the supersonic attempt. The rocket reached 1,367 feet per second, about Mach 1.2, at 2.14 seconds into the flight, and broke apart at about 2.20 seconds — still under thrust, 81 percent of the way through a 2.72 second burn, near maximum dynamic pressure. The avionics bay came down intact, and I was able to pull the complete flight record off it: 500 Hz inertial data and 50 Hz barometric data covering the whole flight.



#### WHAT THE DATA SHOWED

I spent weeks going through that data and the wreckage. The data showed that side loading went from 3 G to 88 G in about a third of a second, the roll rate decayed from about 535 to 290 deg/s, and all four fins were missing from what I recovered.

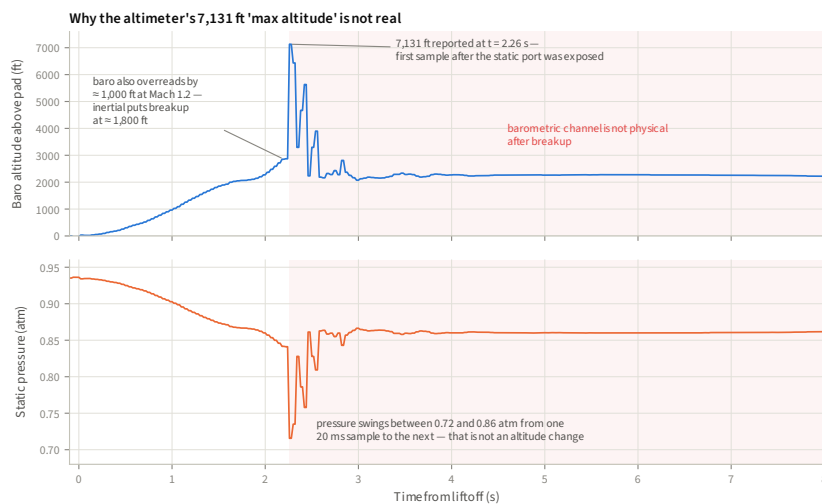
My first read was that the airframe wasn't strong enough — but the roll rate started dropping before the airframe came apart, which meant the fins failed first and the airframe opened afterward. I had to throw out my own explanation and go with what the data showed.



The last 0.7 seconds before rupture, from the 500 Hz record. Roll rate peaks at about 535 deg/s and then decays while the side load grows from 3 G to 88 G. At  $t \approx 2.20$  s both channels go non-physical, which is the airframe letting go rather than any sensor limit.

## THE ALTITUDE THAT WASN'T REAL

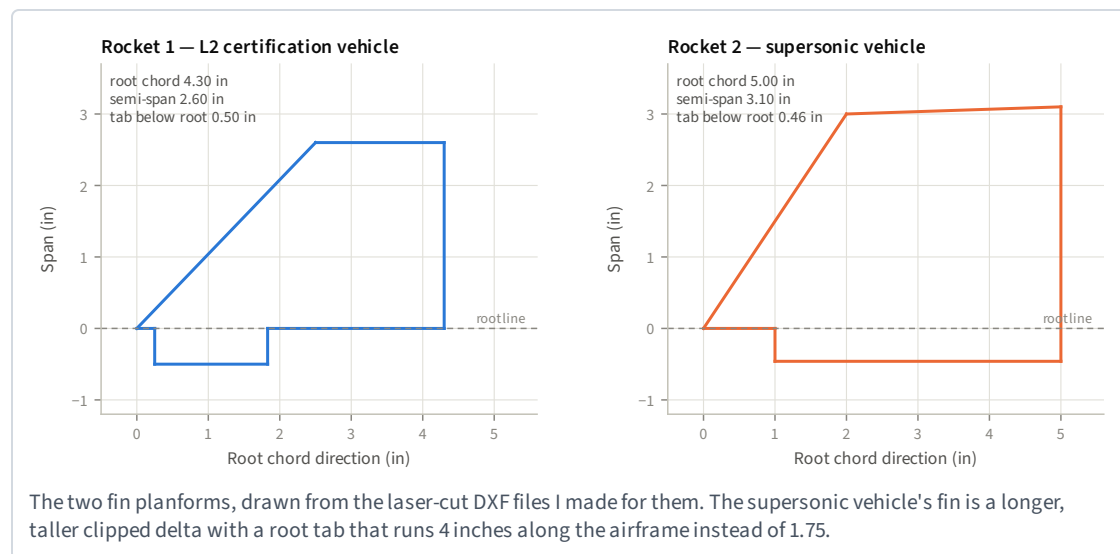
The altimeter's summary file reported a peak altitude of 7,131 feet, far above what actually happened. When the airframe opened, the static port was exposed, and the pressure reading started swinging between 0.72 and 0.86 atm from one 20 ms sample to the next, which is not an altitude change. By cross-checking it against the backup inertial data I was able to establish the real trajectory, and threw the corrupted value out. The barometric channel was not trustworthy before the breakup either: at Mach 1.2 the pressure at a flush static port is not free-stream static, and that channel reads roughly a thousand feet high. Integrating the inertial record instead puts the breakup at about 1,800 feet.



Why the altimeter's 7,131 ft summary value is an artifact: the static pressure channel becomes non-physical the moment the airframe opens.

## WHAT I CHANGED

From there I specified the redesign: thicker, stiffer fins with a stronger root attachment, and a composite airframe replacing the original tube. The changes I ended up making were materials and manufacturing decisions rather than changes to the shape of the rocket. The rebuild is in progress and has not flown yet.

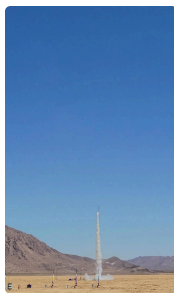


#### WHAT I LEARNED

What has stayed with me from this flight is that the data has the final say. My first explanation was wrong, the number the altimeter handed me was wrong, and the only way to get to the real answer was to go through the record channel by channel and let it correct me. Typically, I design a part, build it, fly it, and use the data from that flight to decide what needs to change for the next version. This flight is the clearest example I have of that loop.



The finished vehicle on 14 July 2026, a month before it flew. 82 inches on a 3.1 inch Blue Tube airframe, with a fiberglass ogive nosecone, four fiberglass fins and a 54 mm motor mount.



The subsonic shakedown leaving the pad, 8 August 2026 — the flight that had to work before I was willing to put the vehicle through Mach 1. It reached 2,833 ft and recovered cleanly. Frame from my own video.



The Eggfinder receiver showing the distance out to the rocket on the lakebed, 8 August 2026. Receiver, transmitter and GPS module are all kits I soldered and assembled, and this is what walking out to a recovery actually looks like.

# Project EXO Bronco – Electronics & Launch Operations co-lead

Two-stage rocket targeting the edge of space · Undergraduate Missiles, Ballistics, and Rocketry Association · sub-team of four · avionics bay wiring, integration and pre-flight checkout

**2**  
stages on the vehicle

**4**  
people on the sub-team I co-lead

**May 2026**  
co-lead since

|        |                                                                                                            |
|--------|------------------------------------------------------------------------------------------------------------|
| ROLE   | Co-lead of the Electronics & Launch Operations sub-team, responsible for the vehicle's avionics.           |
| STATUS | Active · vehicle in development                                                                            |
| TOOLS  | Avionics integration · Wiring harness assembly · Pre-flight checkout · Launch operations · Team leadership |

I co-lead the electronics sub-team for a two-stage rocket aimed at the edge of space. What is mine on this vehicle is the avionics bay wiring and integration, the pre-flight checkout, and the direction the electronics take. The flight software is not mine and I do not claim it.

## THE PROJECT

EXO Bronco is a two-stage rocket that UMBRA, Cal Poly Pomona's student rocketry organization, is developing with the goal of reaching the edge of outer space. I became the co-lead of the electronics sub-division in May 2026, and the sub-team is responsible for the vehicle's avionics and for launch operations.

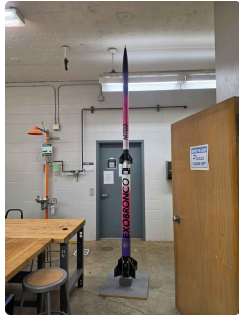
## WHAT I DO

Here, I co-lead a team of 4 people through the design and analysis of the electronics required for a two-stage rocket, which means assigning the work, deciding what has to be finished before each meeting, and setting the direction the electronics take. I have not yet written the electronics requirements themselves. That is the next thing on my list, and I would rather say so than claim a document that does not exist.

On the hardware side I wired and integrated the avionics bay and completed pre-flight checkout and flight readiness preparation. That is the part of this vehicle I have done with my own hands. The custom telemetry board it carries runs flight software written by other members of the organisation, not by me, and I do not claim it. I coordinate with the other sub-teams so that the avionics interfaces match the vehicle-level design decisions.

## WHY IT MATTERS TO ME

My designs and analysis have changed the original design of the electronics I inherited when I first joined. The vehicle has not flown in its current form, so I cannot tell you the design works, only that it is mine and that it is documented well enough for somebody else to build and check. Leading a team like this that's part of a greater project is an incredible feeling.



MTD3 in the Space Lab, the test vehicle carrying the EXOBRONCO name.  
Photograph by Bruce Duong.



Ground test of the recovery system, the two airframe sections still joined by the shock cord after the charge fired. Photograph by Bruce Duong.

Vehicle design details belong to the team and are shown here only with UMBRA's permission.

# Small Rocket Competition – sub-team lead

4-person team · 2,800 ft target altitude · flew 2,514 ft · Adafruit Metro Mini 328 and BMP390 logger on a micro SD breakout · 3D-printed nosecone, boat tail and avionics bay · 29 mm AeroTech G64W

|        |                                                                                                                                                                                                                                                                                  |
|--------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROLE   | Team lead. I ran the schedule and task assignment, reviewed and signed off the CAD before printing, harnessed the avionics bay, and wrote the logger code. A teammate, Eric, did the SolidWorks modelling — I did not know SolidWorks well enough at that point to do it myself. |
| STATUS | Flown · end of Fall 2025                                                                                                                                                                                                                                                         |
| TOOLS  | Arduino C (Metro Mini 328, BMP390, SD logging) · Avionics bay layout and harnessing · Hardware fault isolation · OpenRocket · SolidWorks review and sign-off · FDM 3D printing · Team leadership and scheduling                                                                  |

## 2,514 ft

actual apogee against a 2,800 ft target

## 2nd

where we would have scored, before the disqualification

## 0.25 s

logging interval, time and altitude, to SD

## 4

people on the team

### THE OBJECTIVE

The Small Rocket Competition was my first rocket. UMBRA gave each team a target altitude of 2,800 feet and a common parts list, and the team that came closest won. We were meant to be five people, but one dropped before the design phase, so we finished as four.

### WHAT I DID, AND WHAT I DID NOT

I led the team. That meant running the schedule, deciding what had to be finished before each meeting, assigning the research and design work, and reviewing the SolidWorks files and giving the final go-ahead before anything went to the printer. I want to be accurate about the CAD: Eric did the SolidWorks modelling. I did not know SolidWorks well enough at that point to do it myself, so my part was reviewing and approving it rather than drawing it. Once we reached construction the four of us built the rocket together.

The two things that were mine end to end were the avionics bay and the code. I laid out the bay on a printed sled, harnessed it, held the boards down with rubber bands, and used quarters as ballast to balance the weight of the 9 V battery. The competition supplied the parts — an Adafruit Metro Mini 328 V2, a BMP390 pressure sensor and a micro SD breakout — and I wrote the flight code that ran on them: about eighty lines of Arduino C that read barometric altitude and wrote a time and altitude pair to the SD card every 0.25 seconds, in a format that opened directly in Excel.

### THE ALTIMETER THAT WOULD NOT WORK

My altimeter worked briefly on the day it was handed to us and then stopped. I took it home to fix it and could not get it to read at all. I worked through the wiring against the pinout, moved it from SPI to I2C, ran a bus scan that showed SDA and SCL both sitting high with no device answering, and stripped it back to the sensor alone with the SD breakout removed. The code compiled and ran, and the sensor was simply not there on the bus. I

suspected the hardware rather than the code, but suspecting is not proving. A teammate suggested the test that settled it: run my code on another team's altimeter. It worked first time. That isolated the fault to my hardware, UMBRA replaced the unit, and the replacement flew.

#### WHAT HAPPENED ON THE FLIGHT

The rocket flew on 23 November 2025 and reached 2,514 feet against the 2,800 foot target, measured by a separate altimeter the competition supplied. The logger I built recorded the whole ascent. I no longer have that data file, and the only record of the standings I kept is a photograph of the results slide, but I still have the rocket and the logger boards themselves. On score we were first among the disqualified teams and would have placed second overall.

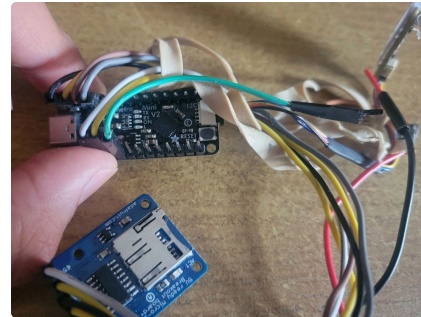
We were disqualified because the motor came out of the rocket. The boat tail was printed in PLA and doubled as the motor retainer, which meant a structural part with a job to do was sitting directly against the hot end of the motor casing. The heat softened it, the ejection charge then pushed the weakened plastic out of the way, and the motor slipped out of the back of the airframe. We found it later by walking the field, and nothing else on the vehicle was damaged.

#### WHAT IT LED TO

That failure is the reason every rocket I have built since uses a metal motor retainer threaded into the aft end, and the reason I treat any part that carries a load and sees motor heat as a structural problem before an aesthetic one. It is also the launch that got me into high power rocketry: by the end of the spring semester I had my Level 1 certification, and by August 2026 my Level 2.



The Small Rocket Competition vehicle at the launch site, 23 November 2025. Printed Haack nosecone, 2.2 in cardboard airframe, four birch plywood fins and a printed conical boat tail at the aft end. Twenty-four inches from tip to tail.



The logger hardware, still on my desk: the Adafruit Metro Mini 328 V2 that ran my code and the micro SD breakout it wrote to, with the harness I made for them.

## SR-30 turbojet performance test

Turbine Technologies SR-30 turbojet · test runs to 70,000 RPM · Excel data reduction · formal technical report

ROLE: Turbojet operator. The test plan splits the crew into six roles and I ran the engine itself, then did the data reduction afterwards.

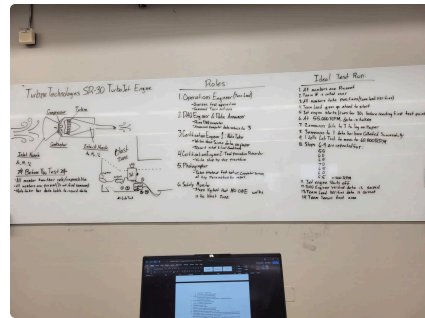
STATUS: Complete · Fall 2025 · TOOLS: Test operation under written procedure · Instrumentation readings · Microsoft Excel data reduction and charting · Technical reporting

### WHAT I DID

The SR-30 is a small turbojet engine on a laboratory test stand. The written test plan divides the crew into six roles and mine was turbojet operator, which means I was the one running the engine. I ran it under the laboratory safety and startup procedures through test runs up to 70,000 RPM, then took the engine performance data into Excel and produced thrust and temperature trend charts for the formal technical report. This is the lab and test-stand side of my experience: running real hardware under a written procedure and turning the instrumentation readings into a report someone else can use.



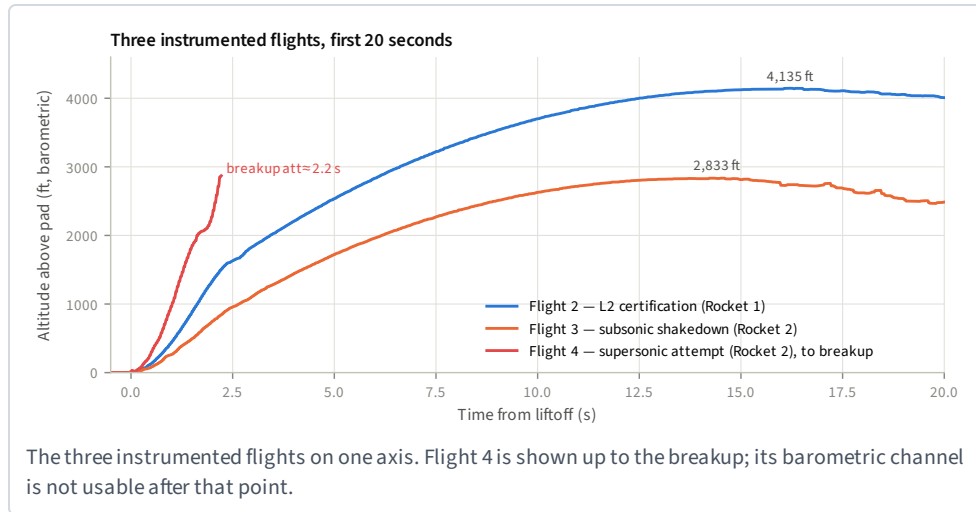
The SR-30 running in its test enclosure, November 2025. Hearing protection and the enclosure are part of the written startup procedure.



The test plan before the run: engine schematic, the six crew roles, and the fourteen-step ideal test run the team worked to.

# Flight log

Four flight tests, two vehicles, two flights each. Two of them failed, and both failures were diagnosed and turned into a redesign — the first one was re-flown and certified, the second one is being rebuilt. The table is on the cover page.



## Now

What I am working on this semester. These are in progress, so they are listed as scope rather than results.

### Rocket 2 rebuild

Thicker, stiffer fins with a stronger root attachment, and a composite airframe replacing the original tube. The changes are materials and manufacturing decisions rather than changes to the shape of the rocket. Not yet flown.

### PIV design team — Design Liaison, since August 2026

I joined the design team for PIV (Particle Image Velocimetry) at Cal Poly Pomona as the design liaison, where I am responsible for creating and maintaining the team's Gantt chart. The main project is an airfoil with tubing at the leading edge that sucks or blows air to keep the flow attached, thinning the boundary layer and keeping it laminar. The goal is to finish the airfoil design by the end of the fall semester and begin testing in the spring.

### ARO 2011L conceptual design project — Team Lead, Fall 2026

I lead a five-person team through a conceptual design of the Grumman F-14A Tomcat for Fundamentals of Systems Engineering & Design, presented as Grumman to NAVAIR in the 1968 VFX era. The deliverables build up into a Conceptual Design Review chart set and then a Preliminary Design Review.

### Fall 2026 coursework

Engineering Statics · Materials Science & Engineering · Aerospace Computational Methods · Fundamentals of Systems Engineering & Design · Engineering Graphics.

# Skills and tools

Each skill is listed with the project it was used on. I have not listed analysis tools I have not used yet; MATLAB and finite element analysis are the next two I am working on.

## CAD & SIMULATION

SolidWorks — part modeling, assemblies, drawings both rockets, electronics bays

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OpenRocket — flight simulation and stability margin Small Rocket Competition, both rockets

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## AVIONICS & FLIGHT TEST

Dual-deploy recovery electronics and deployment logic Blue Raven 2.0 altimeter, Eggfinder GPS tracking

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Ejection charge sizing and pre-flight checkout both rockets, EXO Bronco

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Launch operations NAR Level 2 High Power Rocketry

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## PROGRAMMING

C++ and Java coursework

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Embedded sensor and data-logging code Small Rocket Competition onboard logger

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## FABRICATION & HARDWARE

FDM 3D printing electronics bays for both high power rockets, nosecones, competition prototypes

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Soldering and wiring harness assembly Eggfinder GPS / TX / RX kits, deployment charges, EXO Bronco avionics bay

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## DATA & ANALYSIS

Flight data reduction — 500 Hz inertial and 50 Hz barometric records supersonic failure reconstruction

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Failure investigation from telemetry and wreckage two independent diagnose-and-redesign cycles

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Microsoft Excel data analysis and charting · technical reporting SR-30 turbojet report

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Requirements documentation · Gantt charts EXO Bronco, PIV

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## CERTIFICATIONS

NAR Level 2 High Power Rocketry Certification National Association of Rocketry · August 2026

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NAR Level 1 High Power Rocketry Certification National Association of Rocketry · Spring 2026

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# About

I chose aerospace engineering because it's a place that lives on the frontier of human ingenuity. Here, one mistake can mean the difference between a flight that breaks into new territory and a failure. Speaking generally, I enjoy a good challenge. Coupling that with my interest for space and logical nature, aerospace engineering was the natural path. Specifically, however, I like that “failure” isn't the end of an aerospace engineer's job, as failure can be reconstructed through analysis into a success — and that is something I have firsthand experience with.

Typically, I design a part, build it, fly it, and use the data from that flight to decide what needs to change for the next version. Up to this point, I have been the one setting requirements and deciding when the project was good enough to proceed. Thus far I have been reasoning from test data, intuition, and analysis I could find online, and I would like to be able to predict how a system will react before I build it, then test how the actual system works. Ultimately, I want to work on a project greater than myself — a project that I can help build up and learn how to be a proper engineer in the process.

Alongside school I have worked as a gurney and wheelchair transport technician for Transport Today since June 2024, moving non-emergency patients between hospitals, clinics, and care facilities and following the safety procedure through loading, transit, and unloading.

|               |                                                                                                                     |
|---------------|---------------------------------------------------------------------------------------------------------------------|
| SCHOOL        | California State Polytechnic University, Pomona                                                                     |
| DEGREE        | B.S. Aerospace Engineering · Expected August 2029                                                                   |
| STANDING      | GPA 3.97 · President's Honor List, 2025–2026                                                                        |
| ORGANIZATIONS | UMBRA (Undergraduate Missiles, Ballistics, and Rocketry Association) · PIV (Particle Image Velocimetry) design team |
| COURSEWORK    | Calculus I–III · Advanced Engineering Mathematics · Physics I (Mechanics) · General Chemistry                       |
| STATUS        | U.S. citizen · eligible for a U.S. government security clearance                                                    |
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